

TRB-009 — Discovery

Submitter: **Test Submitter**

Request: *Submit button stuck on spinner — submissions failing with 502*

Proposed Solution: TRB-009 — Submit button stuck on spinner

What is proposed

- Add server-side health and idempotency checks in `/api/submit`, returning a structured error envelope on 5xx instead of silently failing the response.
- Wrap the submit handler in the rebate form with a retry-with-backoff (max 2 retries, 8s total) and a circuit-breaker that surfaces a user-visible error after the cap.
- Add front-end error UI: replace the indefinite spinner state with either "Submitted" or a retry / contact-support card.
- Add structured logging on the submit path so 502s are linked to a request id the user can quote when contacting support.

Desired outcome

- A user who clicks Submit reaches one of three deterministic states within 15 seconds:
 1. confirmation screen, OR
 2. an actionable "we couldn't submit" message with a request id and retry control, OR
 3. an offline / unreachable card if the network is down.
- No production user is ever left staring at an indefinite spinner.
- 502 spikes from the upstream service are visible in the dashboard within 60 seconds and paged if >2% of submissions fail for 5 minutes.

Success criteria

- Synthetic test in CI simulates a 502 from `/api/submit` and asserts the UI renders the retry card within 15 seconds.
- Production telemetry for `arkansas\_rebate\_submit\_success\_rate` reports $\geq 99.5\%$ over a 7-day window.
- p95 client-observed submit latency ≤ 8 seconds; p99 ≤ 15 seconds.
- Manual smoke from the staging Arkansas-Tax-Rebate-Portal: log in as a test client, fill the rebate form, click Submit, observe the confirmation screen.